Power BI Module End Assignment

Task:

| Trip_ID | Vehicle_ID | Driver_ID | Origin | Destination | Distance_km | Fuel_Consumed_L | Delivery_Status | Delivery_Date |
|---|---|---|---|---|---|---|---|---|
| T001 | V04 | D01 | Delhi | Pune | 1173 | 108.42 | On-Time | 1/27/2023 |
| T002 | V06 | D08 | Mumbai | Bangalore | 1727 | 161.33 | On-Time | 2/21/2023 |
| T003 | V06 | D08 | Mumbai | Pune | 1459 | 154.7 | On-Time | 2/17/2023 |
| T004 | V04 | D09 | Hyderabad | Pune | 382 | 26.6 | On-Time | 2/18/2023 |
| T005 | V06 | D08 | Pune | Mumbai | 398 | 33.2 | On-Time | 2/15/2023 |
| T006 | V06 | D07 | Chennai | Mumbai | 1275 | 85.04 | Late | 2/25/2023 |
| T007 | V07 | D03 | Chennai | Kolkata | 752 | 58.08 | On-Time | 1/19/2023 |
| T008 | V02 | D10 | Delhi | Pune | 74 | 5.24 | On-Time | 1/1/2023 |
| T009 | V02 | D07 | Delhi | Hyderabad | 186 | 16.22 | On-Time | 2/23/2023 |
| T010 | V02 | D02 | Bangalore | Hyderabad | 1375 | 105.21 | Late | 2/2/2023 |
| T011 | V06 | D03 | Kolkata | Hyderabad | 419 | 31.17 | On-Time | 1/21/2023 |
| T012 | V06 | D01 | Kolkata | Delhi | 751 | 51.77 | On-Time | 2/15/2023 |
| T013 | V05 | D04 | Kolkata | Chennai | 1571 | 188.52 | Late | 2/2/2023 |
| T014 | V05 | D05 | Hyderabad | Bangalore | 1524 | 104.51 | On-Time | 2/16/2023 |
| T015 | V05 | D06 | Kolkata | Mumbai | 1956 | 179.88 | On-Time | 1/21/2023 |
| T016 | V05 | D06 | Bangalore | Mumbai | 858 |  | Late | 1/18/2023 |
| T017 | V07 | D07 | Pune | Kolkata | 1269 | 102.91 | On-Time | 1/16/2023 |

Cleaned the dataset and Identify the missing values

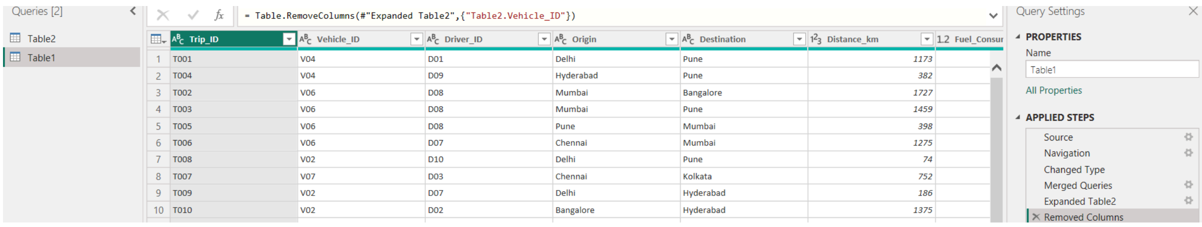
Replace Missing Values using the Mean method in Excel.

Correct the inconsistent date formats to ensure data consistency.

| Trip_ID | Vehicle_ID | Driver_I | Origin | Destination | Distance_km | Fuel_Consumed_L | Delivery_Status | Delivery_Date |
|---|---|---|---|---|---|---|---|---|
| T001 | V04 | D01 | Delhi | Pune | 1173 | 108.42 | On-Time | 27-01-2023 |
| T002 | V06 | D08 | Mumbai | Bangalore | 1727 | 161.33 | On-Time | 21-02-2023 |
| T003 | V06 | D08 | Mumbai | Pune | 1459 | 154.70 | On-Time | 17-02-2023 |
| T004 | V04 | D09 | Hyderabad | Pune | 382 | 26.60 | On-Time | 18-02-2023 |
| T005 | V06 | D08 | Pune | Mumbai | 398 | 33.20 | On-Time | 15-02-2023 |
| T006 | V06 | D07 | Chennai | Mumbai | 1275 | 85.04 | Late | 25-02-2023 |
| T007 | V07 | D03 | Chennai | Kolkata | 752 | 58.08 | On-Time | 19-01-2023 |
| T008 | V02 | D10 | Delhi | Pune | 74 | 5.24 | On-Time | 01-01-2023 |
| T009 | V02 | D07 | Delhi | Hyderabad | 186 | 16.22 | On-Time | 23-02-2023 |
| T010 | V02 | D02 | Bangalore | Hyderabad | 1375 | 105.21 | Late | 02-02-2023 |
| T011 | V06 | D03 | Kolkata | Hyderabad | 419 | 31.17 | On-Time | 21-01-2023 |
| T012 | V06 | D01 | Kolkata | Delhi | 751 | 51.77 | On-Time | 15-02-2023 |
| T013 | V05 | D04 | Kolkata | Chennai | 1571 | 188.52 | Late | 02-02-2023 |
| T014 | V05 | D05 | Hyderabad | Bangalore | 1524 | 104.51 | On-Time | 16-02-2023 |
| T015 | V05 | D06 | Kolkata | Mumbai | 1956 | 179.88 | On-Time | 21-01-2023 |
| T016 | V05 | D06 | Bangalore | Mumbai | 858 | 60.77 | Late | 18-01-2023 |
| T017 | V07 | D07 | Pune | Kolkata | 1269 | 102.91 | On-Time | 16-01-2023 |

Import Clean dataset into Power bi using Power Query.

Merged Two Tables convert into one table using Power Query for analysis.




| | Trip_ID | Vehicle_ID | Driver_ID | Origin | Destination | Distance_km |
|---|---|---|---|---|---|---|
| 1 | T001 | V04 | D01 | Delhi | Pune | 1173 |
| 2 | T004 | V04 | D09 | Hyderabad | Pune | 382 |
| 3 | T002 | V06 | D08 | Mumbai | Bangalore | 1727 |
| 4 | T003 | V06 | D08 | Mumbai | Pune | 1459 |
| 5 | T005 | V06 | D08 | Pune | Mumbai | 398 |
| 6 | T006 | V06 | D07 | Chennai | Mumbai | 1275 |
| 7 | T008 | V02 | D10 | Delhi | Pune | 74 |
| 8 | T007 | V07 | D03 | Chennai | Kolkata | 752 |
| 9 | T009 | V02 | D07 | Delhi | Hyderabad | 186 |
| 10 | T010 | V02 | D02 | Bangalore | Hyderabad | 1375 |

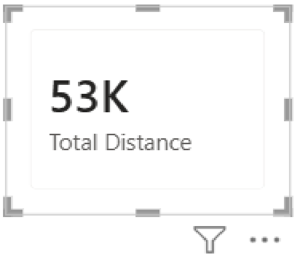
Created Calculate Measure using DAX

```
1  Total Trips = count(Table1[Trip_ID])
```

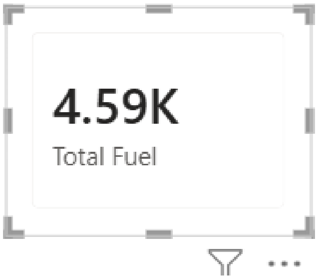
**50**

Total Trips

```
Total Distance = sum(Table1[Distance_km])
```




```
Total Fuel = sum(Table1[Fuel_Consumed_L])
```




```
On-Time Delivers = CALCULATE(COUNT(Table1[Trip_ID]),Table1[Delivery_Status]="On-Time")
```


```
Late Delivers = CALCULATE(COUNT(Table1[Trip_ID]),Table1[Delivery_Status]="Late")
```

Created Calculated Columns to evaluate order performance based on delivery status.

```
1 performance = IF(Table1[Delivery_Status]="On-Time","Good","Needs Improvement")
```

| n | Destination | Distance_km | Fuel_Consumed_L | Delivery_Status | Delivery_Date | Table2.Vehicle_Type | Table2.Capacity_kg | Table2.Maintenance_Cost | performance |
|---|---|---|---|---|---|---|---|---|---|
| | Pune | 1173 | 108.42 | On-Time | 27/01/2023 | Mini-Truck | 8803 | 9033 | Good |
| abad | Pune | 382 | 26.6 | On-Time | 18/02/2023 | Mini-Truck | 8803 | 9033 | Good |
| ɔai | Bangalore | 1727 | 161.33 | On-Time | 21/02/2023 | Van | 6053 | 5914 | Good |
| ɔai | Pune | 1459 | 154.7 | On-Time | 17/02/2023 | Van | 6053 | 5914 | Good |
| | Mumbai | 398 | 33.2 | On-Time | 15/02/2023 | Van | 6053 | 5914 | Good |
| ıai | Mumbai | 1275 | 85.04 | Late | 25/02/2023 | Van | 6053 | 5914 | Needs Improvement |
| | Pune | 74 | 5.24 | On-Time | 01/01/2023 | Truck | 1970 | 6776 | Good |
| ıai | Kolkata | 752 | 58.08 | On-Time | 19/01/2023 | Van | 2598 | 12837 | Good |
| | Hyderabad | 186 | 16.22 | On-Time | 23/02/2023 | Truck | 1970 | 6776 | Good |
| ılore | Hyderabad | 1375 | 105.21 | Late | 02/02/2023 | Truck | 1970 | 6776 | Needs Improvement |
| :a | Hyderabad | 419 | 31.17 | On-Time | 21/01/2023 | Van | 6053 | 5914 | Good |
| :a | Delhi | 751 | 51.77 | On-Time | 15/02/2023 | Van | 6053 | 5914 | Good |
| :a | Chennai | 1571 | 188.52 | Late | 02/02/2023 | Truck | 9941 | 17751 | Needs Improvement |

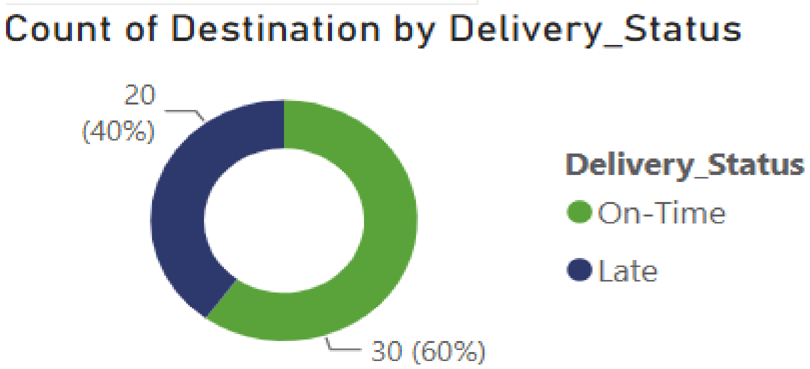
Added New Month Column using with Calculated Column to evaluate month-wise analysis.

```
1 Month = FORMAT(Table1[Delivery_Date].[Month],"MMM")
```

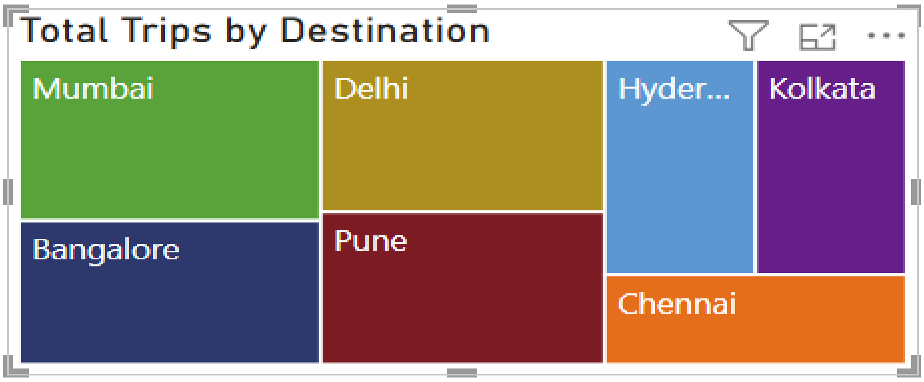
| ıation | Distance_km | Fuel_Consumed_L | Delivery_Status | Delivery_Date | Table2.Vehicle_Type | Table2.Capacity_kg | Table2.Maintenance_Cost | performance | Month |
|---|---|---|---|---|---|---|---|---|---|
| | 1173 | 108.42 | On-Time | 27/01/2023 | Mini-Truck | 8803 | 9033 | Good | January |
| | 382 | 26.6 | On-Time | 18/02/2023 | Mini-Truck | 8803 | 9033 | Good | February |
| ore | 1727 | 161.33 | On-Time | 21/02/2023 | Van | 6053 | 5914 | Good | February |
| | 1459 | 154.7 | On-Time | 17/02/2023 | Van | 6053 | 5914 | Good | February |
| ai | 398 | 33.2 | On-Time | 15/02/2023 | Van | 6053 | 5914 | Good | February |
| ai | 1275 | 85.04 | Late | 25/02/2023 | Van | 6053 | 5914 | Needs Improvement | February |
| | 74 | 5.24 | On-Time | 01/01/2023 | Truck | 1970 | 6776 | Good | January |
| ɜ | 752 | 58.08 | On-Time | 19/01/2023 | Van | 2598 | 12837 | Good | January |
| ıbad | 186 | 16.22 | On-Time | 23/02/2023 | Truck | 1970 | 6776 | Good | February |
| ıbad | 1375 | 105.21 | Late | 02/02/2023 | Truck | 1970 | 6776 | Needs Improvement | February |
| ıbad | 419 | 31.17 | On-Time | 21/01/2023 | Van | 6053 | 5914 | Good | January |
| | 751 | 51.77 | On-Time | 15/02/2023 | Van | 6053 | 5914 | Good | February |
| ai | 1571 | 188.52 | Late | 02/02/2023 | Truck | 9941 | 17751 | Needs Improvement | February |
| ore | 1524 | 104.51 | On-Time | 16/02/2023 | Truck | 9941 | 17751 | Good | February |
| ai | 1956 | 179.88 | On-Time | 21/01/2023 | Truck | 9941 | 17751 | Good | January |

Create DAX Measures and Columns to perform business calculation and generate insights.

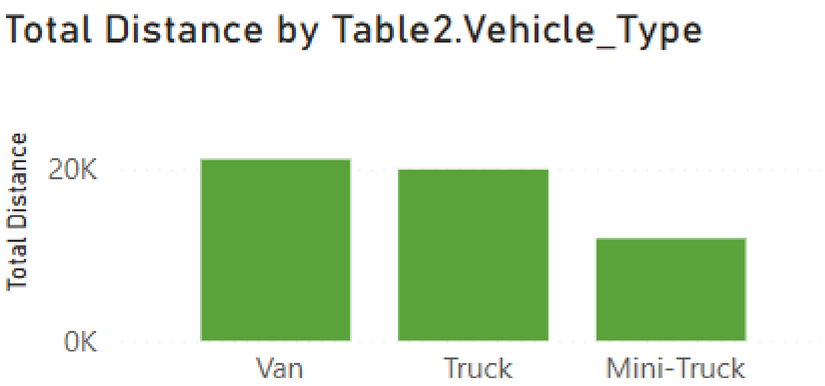
Designed a donut chart to visualize destination counts by delivery status.



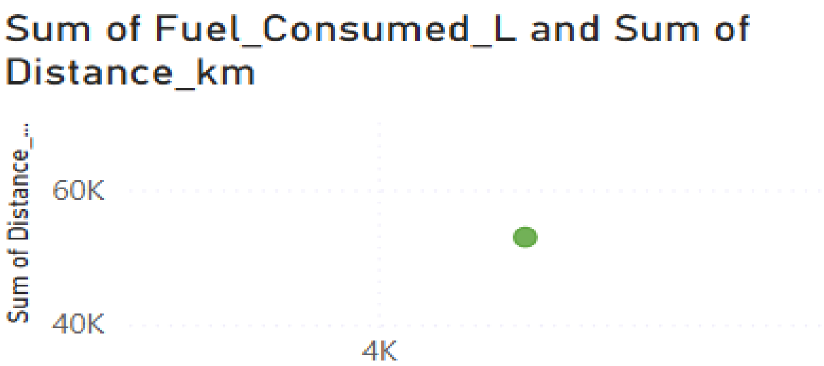
Designed a TreeMap Visualization to analyze the total number of trips by destination.



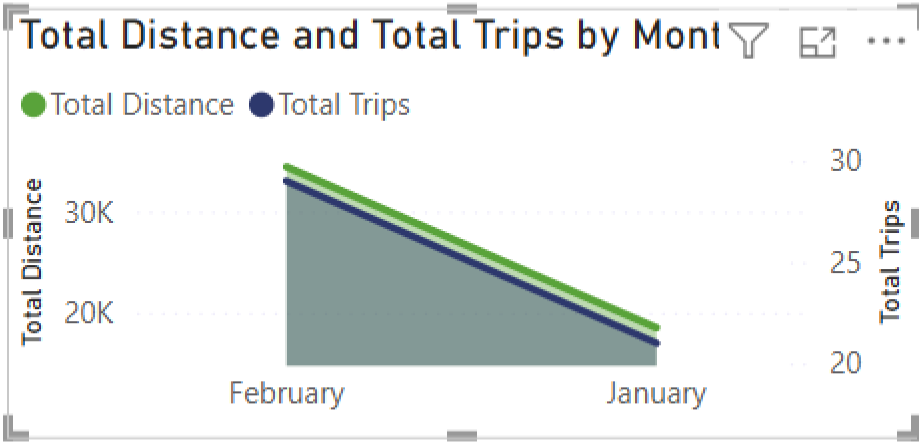

Created a Clustered Column Chart Visualize to Compare the total distance across different vehicle types.



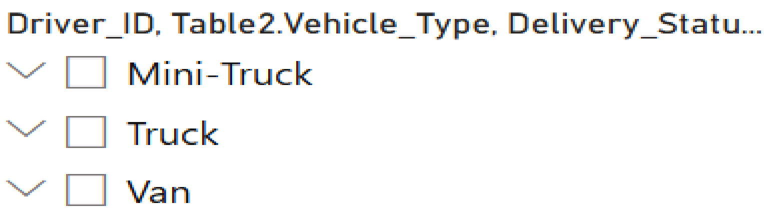

Designed a Scatter Plot Visual to analyze the relationship between fuel consumption and distance traveled.



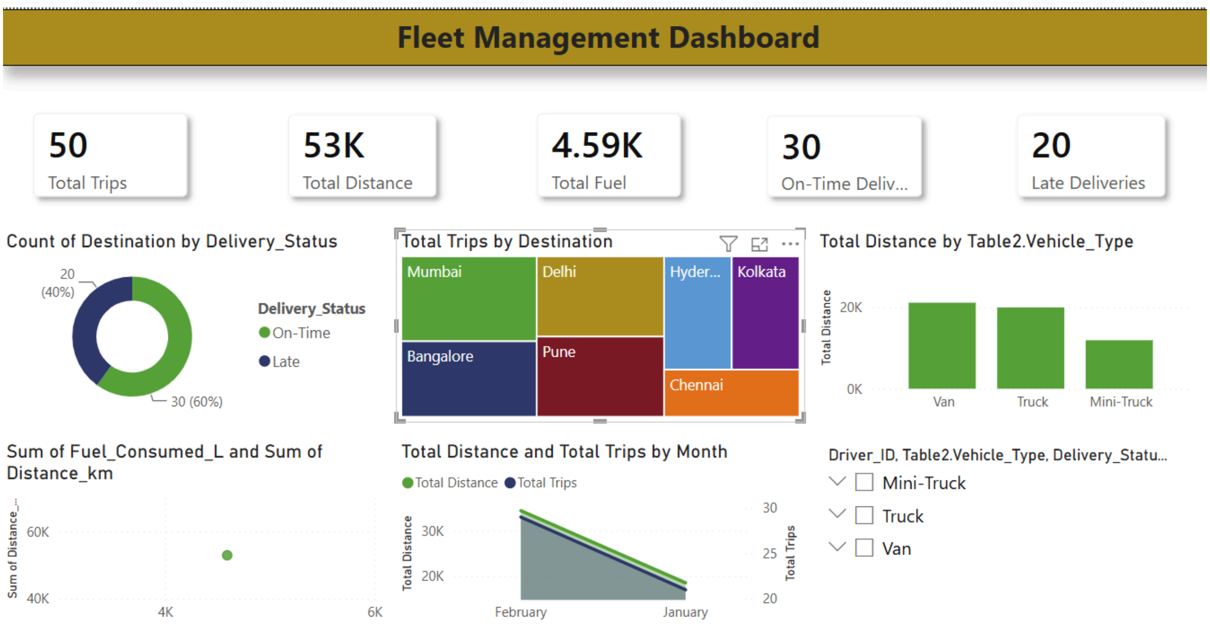
Created an Area Chart visualising the total distance and total trips by  months.




Designed a Slicer to filter data by driver id, vehicle type and delivery status.




Developed an interactive Fleet Management Dashboard to monitor transportation performance and delivery operations.



**Key Insights**

**1. Vans Covered the highest total distance compared Trucks and Mini-Trucks.**
**2. Fuel consumption increased with the distance traveled, indicating a positive relationship between the two variables.**
**3. February records higher total trips and total distance than January.**
**4. 60% of deliveries were completed on-time, while 40% experienced delays.**
**5. Vehicle type and delivery status significantly influenced overall fleet performance.**

**Conclusion**

**The Fleet Management Dashboard provides valuable insights into transportation operations by analyzing trips, distance traveled, fuel consumption, vehicle performance and delivery status. The Dashboard enables effective monitoring of fleet activities, supports data-driven decision making and helps identify opportunities to improve efficiency and delivery performance.**